Examiner only

7. Tom competes in a 400 m training race. His split times to travel each 100 m are measured. His pulse rate is also measured in beats per minute (bpm) before the race starts and at every 100 m interval. The measurements are shown in **Table 1** below.

Table 1

Distance (m)	0	100	200	300	400
Time (s)	0	13	27	43	58
Pulse rate (bpm)	78	133	132	134	133

Use the data in the table and your knowledge to answer the following questions.

(a) (i) State Tom's resting pulse rate. .... bpm [1]

(ii) One minute after the race ended, Tom's pulse rate had dropped to 122.

Calculate **how much longer** it will be for his pulse to return to his resting rate, assuming it continues to drop at the same rate. [3]

time = ... minutes

(iii) Use the equation

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

and the information in **Table 1** to calculate Tom's mean speed over the race. [2]

mean speed = ... m/s

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(iv) Use the information in **Table 1** to complete **Table 2** to show the order of running each 100 m section from fastest to slowest. The fastest 100 m section has already been filled in. [2]

Table 2

Fastest section	0 to 100 m
↓	.... to .... m
	.... to .... m
Slowest section	.... to .... m

(b) Tom then trains for 6 months. **Complete Table 3** below by estimating Tom's pulse rate every 100 m in a 400 m race after training. [2]

Table 3

Distance (m)	0	100	200	300	400
Pulse rate (bpm)	....	....	....	....	....

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